# Where does Qwen3-32B *hold* a paper's authors?

A layer-by-layer readout of author names for 9,108 papers

**Stages 10–11 · jlens**

Qwen3-32B · 2026-10-27

# The question

**target (4,621 papers)**
can't write the authors
but picks the true list 4/4 times

**control (2,220 papers)**
can't write the authors
and can't pick the true list

If the model already **knows** target authors in its middle layers and only fails to say them, a lens on those layers should read the right name **better for target than control**.

If the knowledge is only assembled at the end, the gap should appear only near the output.

**So: at which layer does target pull ahead of control?**

# The data

**9,108**
papers, 8 fields

**33,811**
author names

**149,999**
name tokens scored

**13**
citation bins (0 to 63,901)

**Source:** the **Dreaming Up Authors** evaluation set. 9,999,863 random OAG v2 records + Semantic Scholar metadata (Sep 2025) -> quality filters -> 1.92 M papers -> up to 100 per field x citation bin

**Groups** (group lead's labelling run, same model)

target 4,621 · control 2,220

guard (writes the authors) 214 · excluded 2,053

Medicine, biology, CS, psychology, chemistry, maths, physics, materials science: ~1,100 papers each

**Name formats**

73% start with an initial ("S. Bree")

26% start with a full given name ("Yan Zhao")

# How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

**open answer -> R0 / R1 / R2**

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

**multiple choice -> target / control**

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

**5 false titles per paper**

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier

**statistics**

target vs control only within citation bin

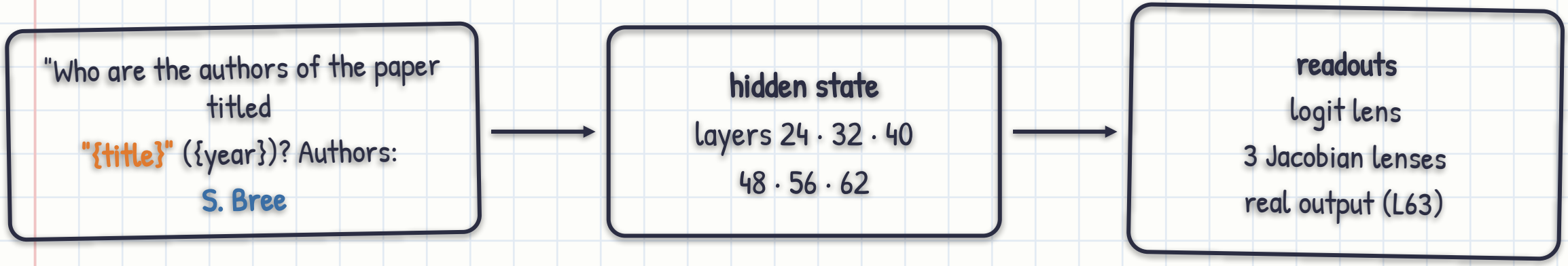
10 bins with >= 30 per group

weight n_t n_c / (n_t + n_c)

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

# How we read the model




**Score** = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

**7 title conditions**: the true title, 5 matched false titles, no title

**title-specific signal**

= true title - mean(5 false titles)

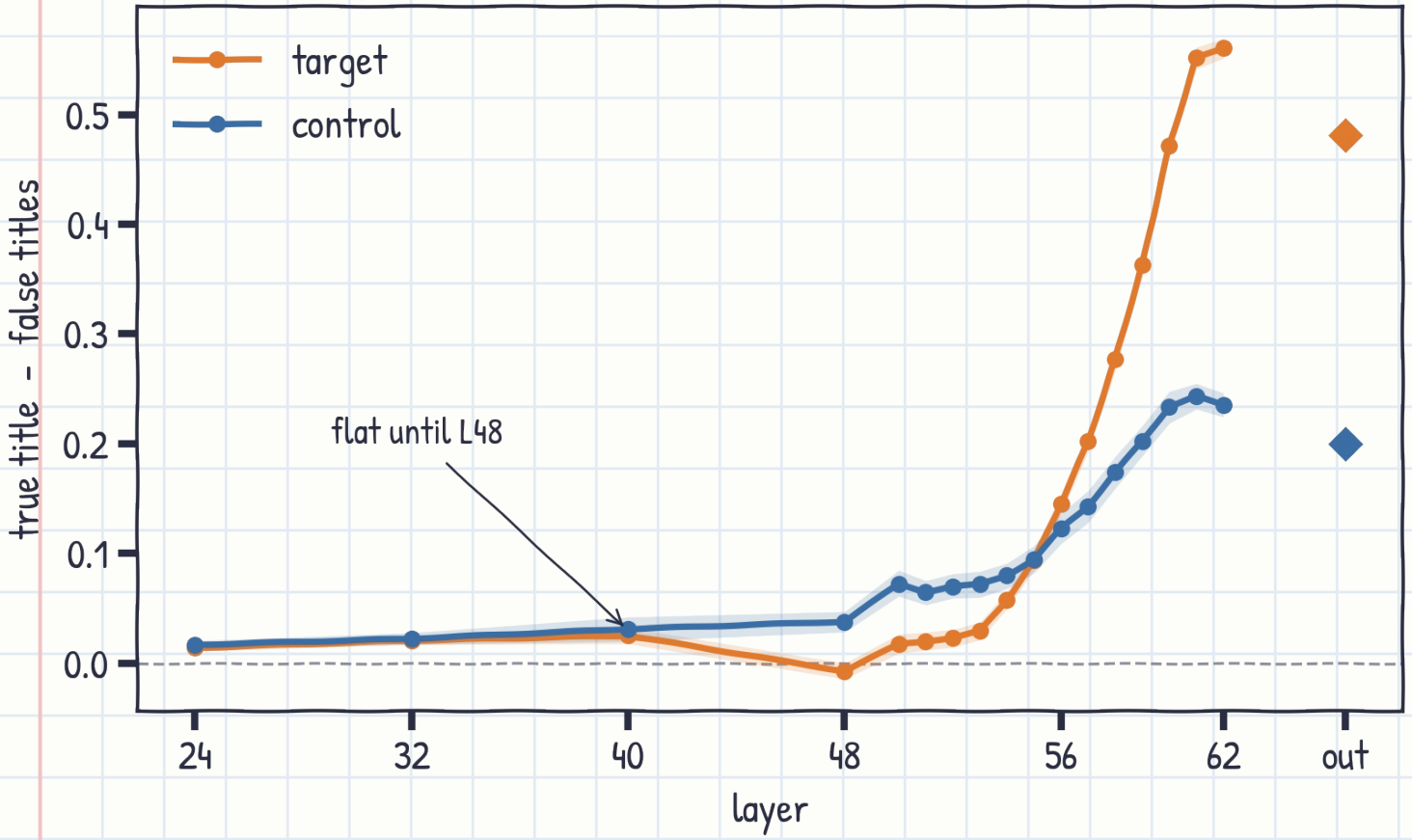
Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid)

# Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



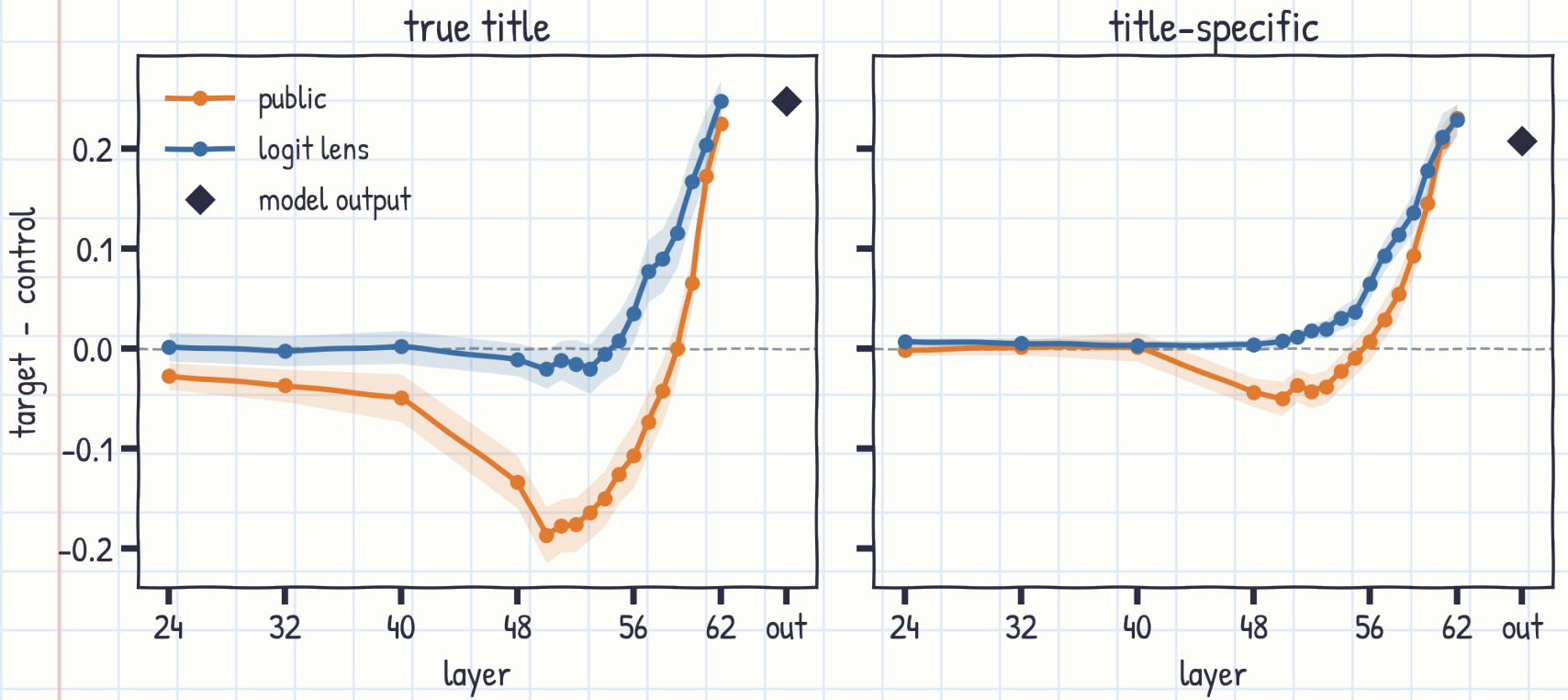

**L24–L48:** ~0 for both groups

rises at **L56**

L62: target +0.56 vs control +0.24

# Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI




Middle layers: **no head start**

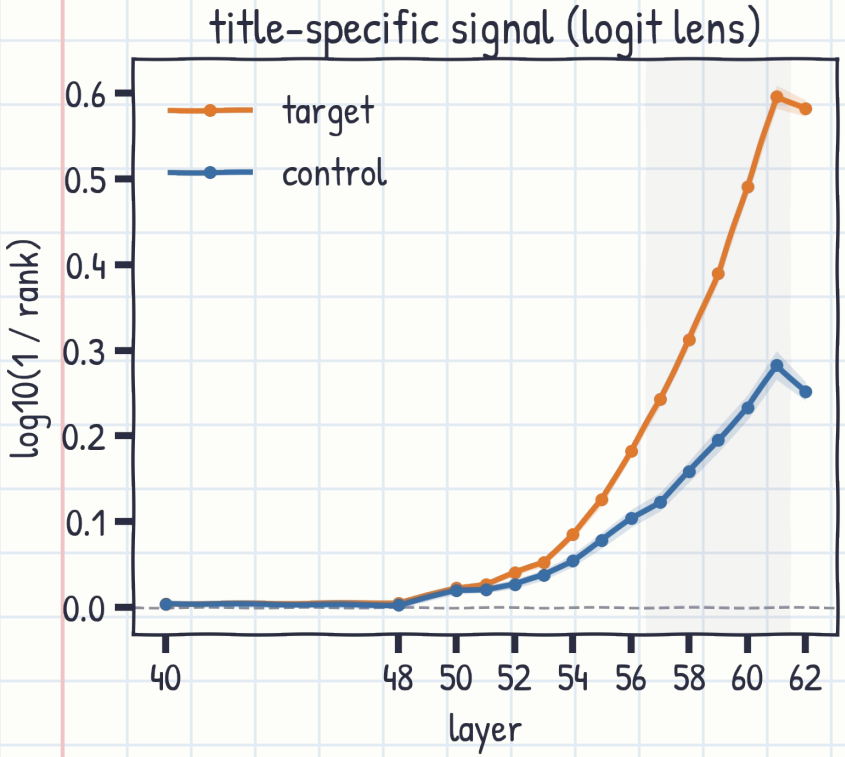
L62 / output:

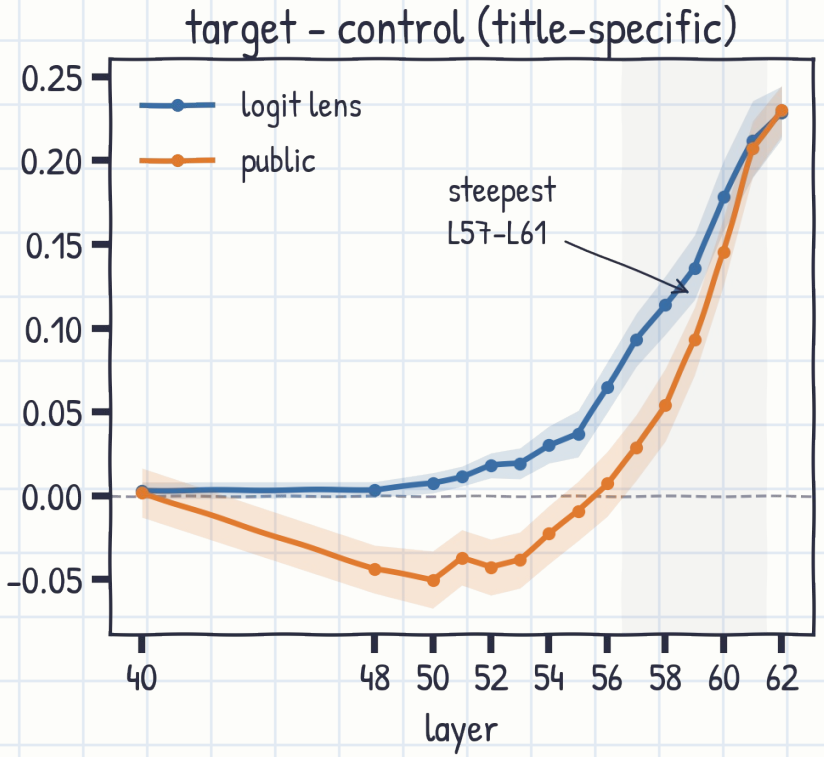
+0.21 to +0.23

almost all title-specific

# It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI




Logit lens: target - control

+0.007 at L50

+0.23 at L62

steepest rise **L57–L61**

# Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 · patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain

**patch**

copy the true-title hidden state into the false-title run at one layer

**knock out**

block attention to the title tokens (all heads) from layer l to the last layer

**probe**

linear map from the last title token to the output state

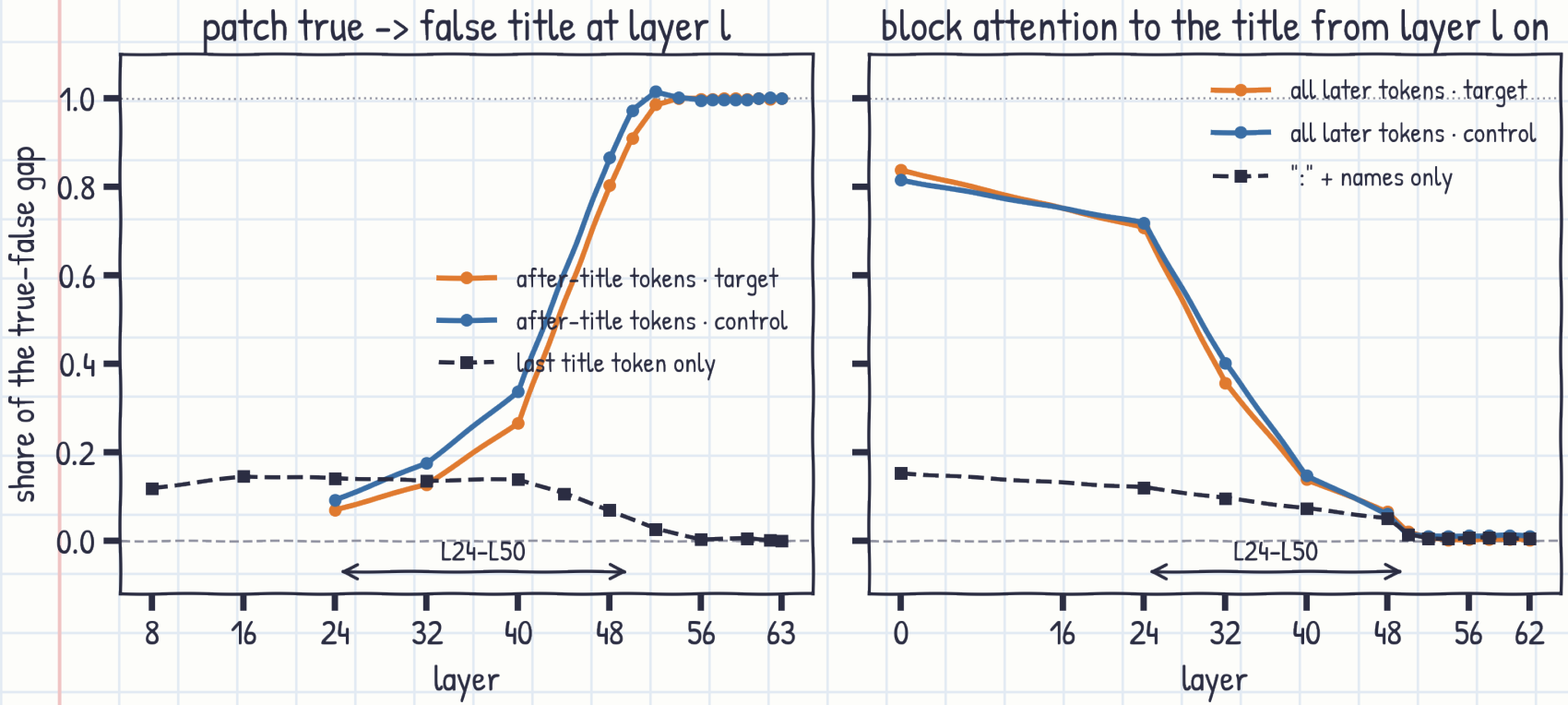
**chain**

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened − false) / (true − false).

# The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02



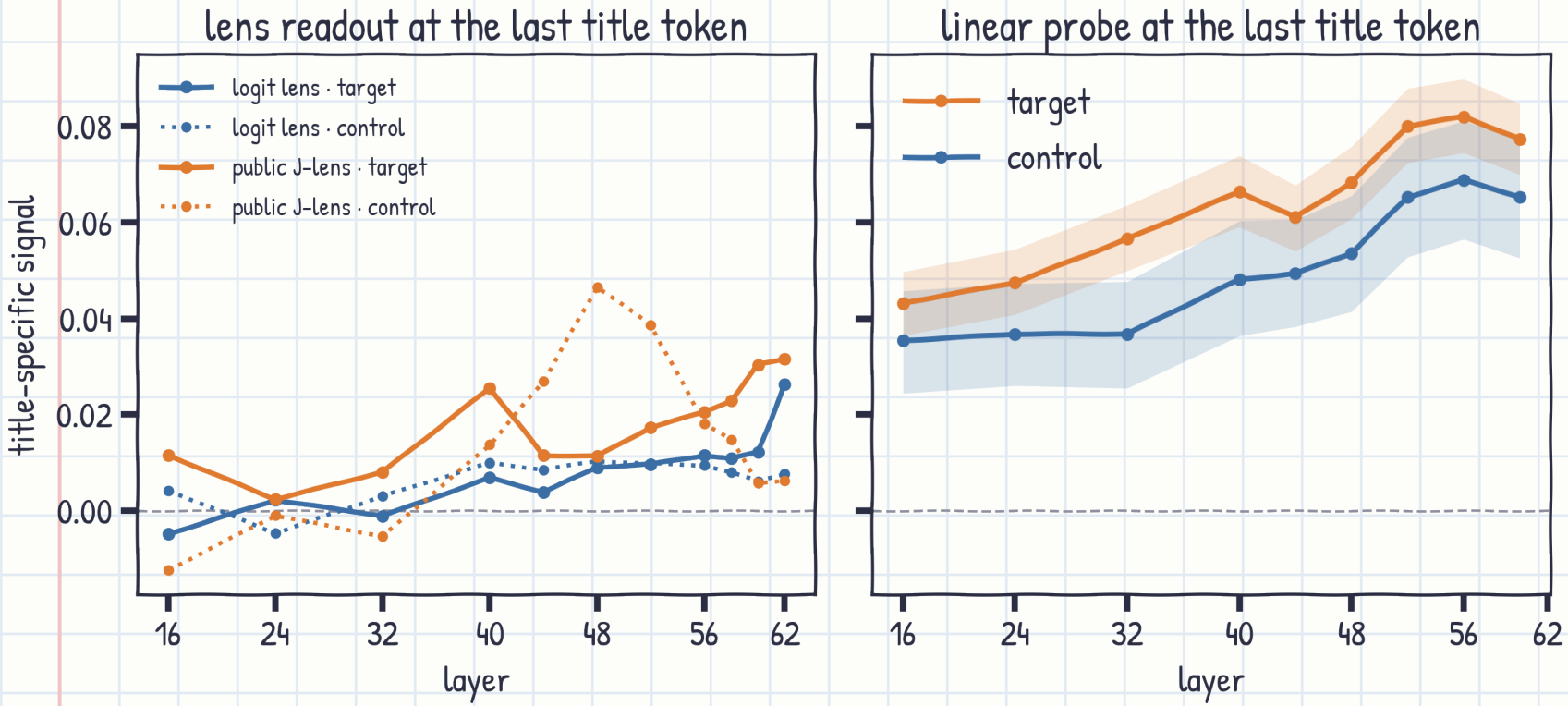

L52+: title no longer needed

info moved to the tokens **right after** the title

same timing for target and control

# Read early: a probe finds it at the title

title-specific signal at the last title token · all 9,108 papers




lenses: ~0.01–0.03

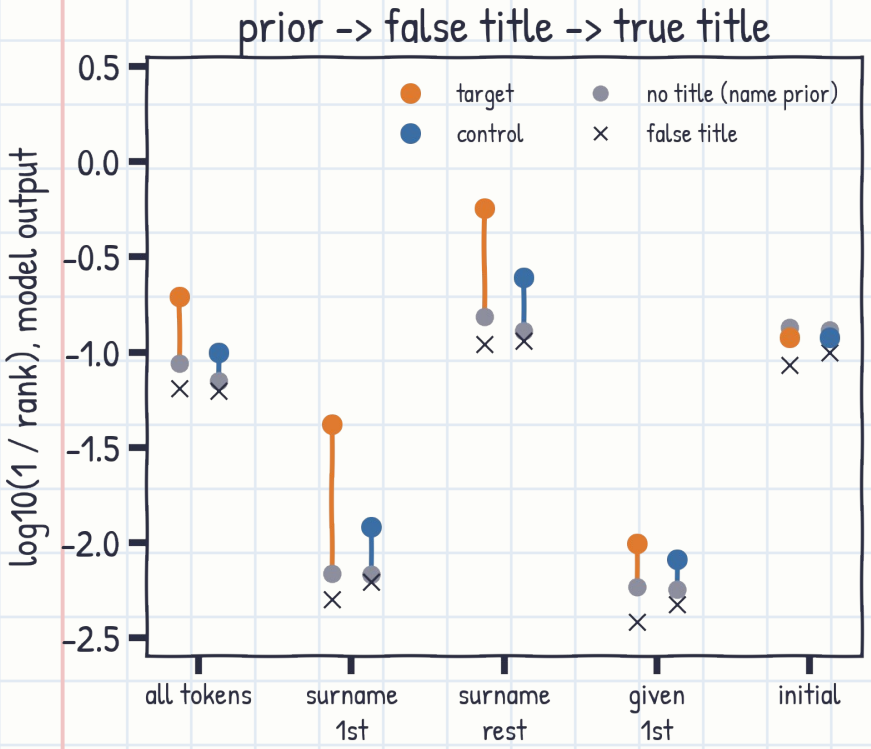
probe: 0.04 -> 0.08

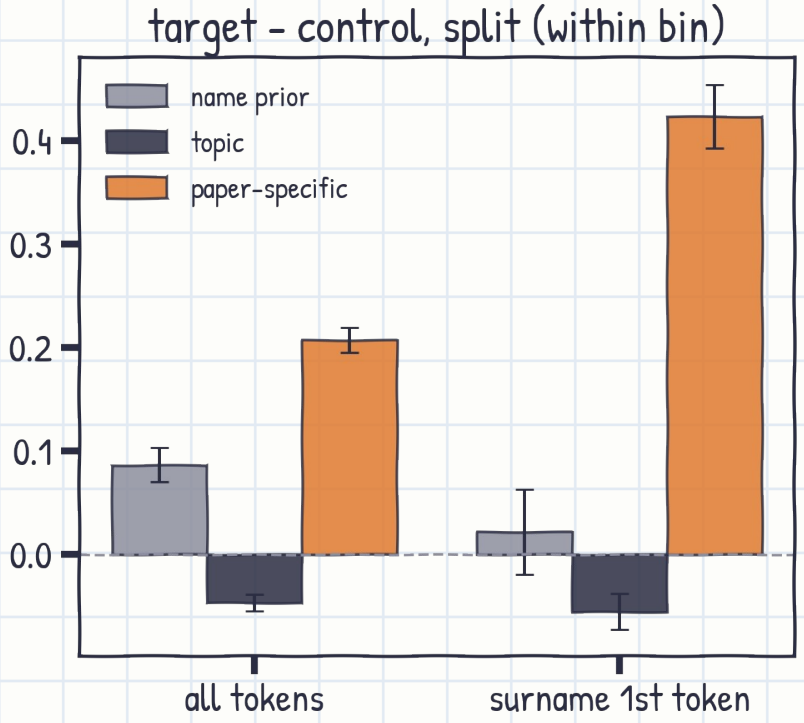
target > control by +0.02–0.04

read early (title, L16–24), spoken late (names, L57–61)

# How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers




no title: rank ~12

target - control: +0.09 prior, +0.21 paper

surname 1st: prior ~0, paper +0.42

prior vs recognition rho **0.09**

# What we learned: read early, spoken late

| | |
|---|---|
| Read early | Causally the title is read in L24–L50 and moved into the tokens after it; after L52 it is no longer needed. A probe at the title token reads a little of it from L16–L24. |
| Spoken late | At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61. |
| No lens lead | Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (probe at the title: +0.02–0.04). |
| Real knowledge | Name prior is ~1/3 of the gap overall but ~0 on surnames; the rest is paper-specific and grows with citations. |
| Lenses agree | Jacobian lenses read more in middle layers, but nothing target-specific. |
| Name format | Initials carry little; full given names carry as much as surnames and help predict them. |

# Caveats and next steps

## Caveats

- Readout is correlational; patching/knockout fix the timing, not the mechanism inside L24–L50
- Parity tolerance raised 0.25 -> 0.5 after the fact
- Groups are model-specific; low-citation "recognition" is near chance
- Few control papers above 1,000 citations
- Label run settings (max length, thinking mode) not in the package

## Stage 12 (running)

1. Patch single positions (quote, year, "Authors:") – done, analysing
2. Attention vs MLP patching in L8–L56
3. A lens fitted on author prompts, read at L24–L62
4. Name-prior baselines: distractor lists without a title; a prompt with no paper

# Appendix

checks · readout levels · author chain · probe controls · lenses · citations · name formats

# Can we trust the numbers?

**fast = slow**

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical

**matches Stage 6**

972 tokens: same token ids, median log-prob gap ~1e-5

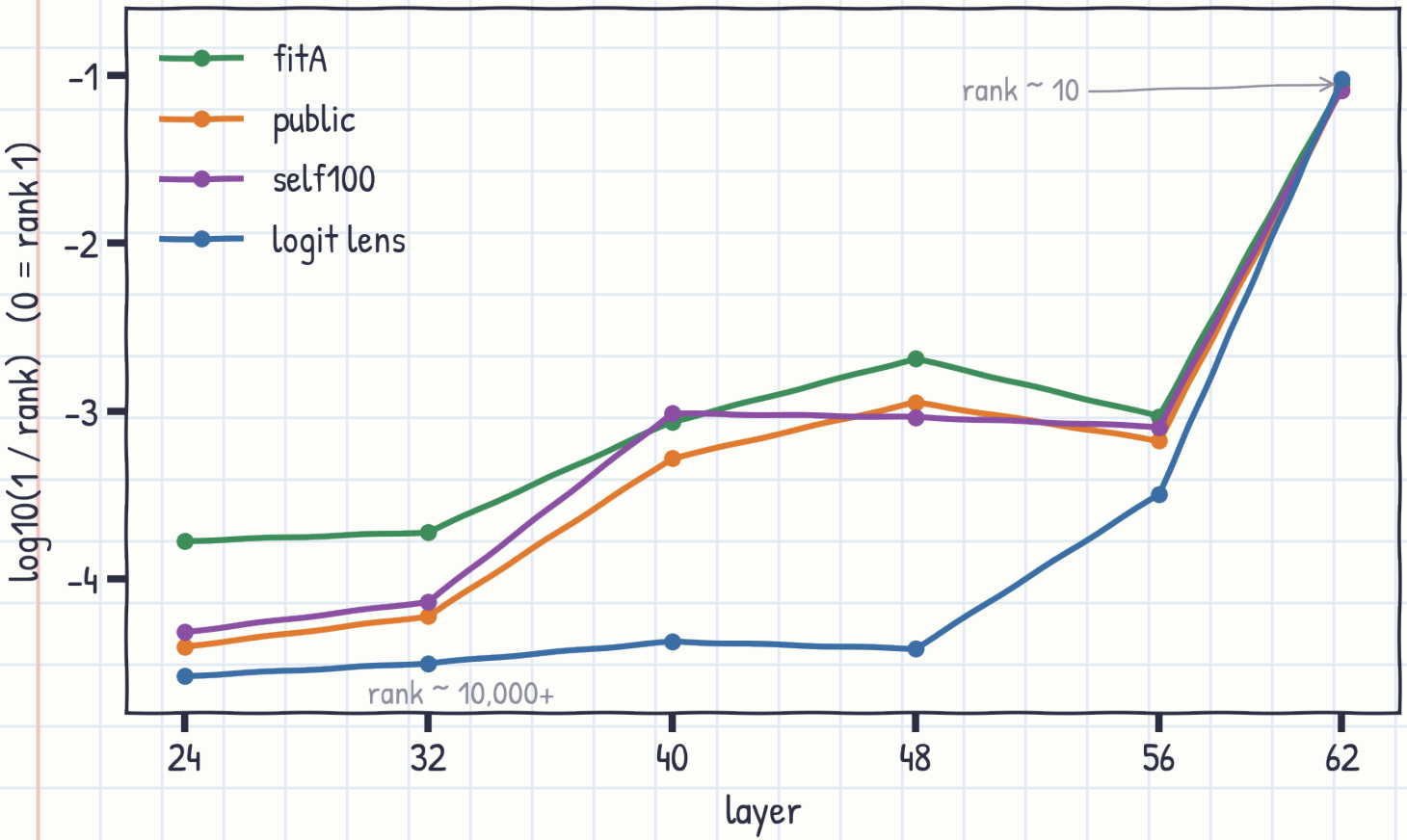
**negative control**

Punctuation tokens show ~0 title-specific signal at every layer

**Caveat:** the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps.

# Names become readable late

all 9,108 papers · true title



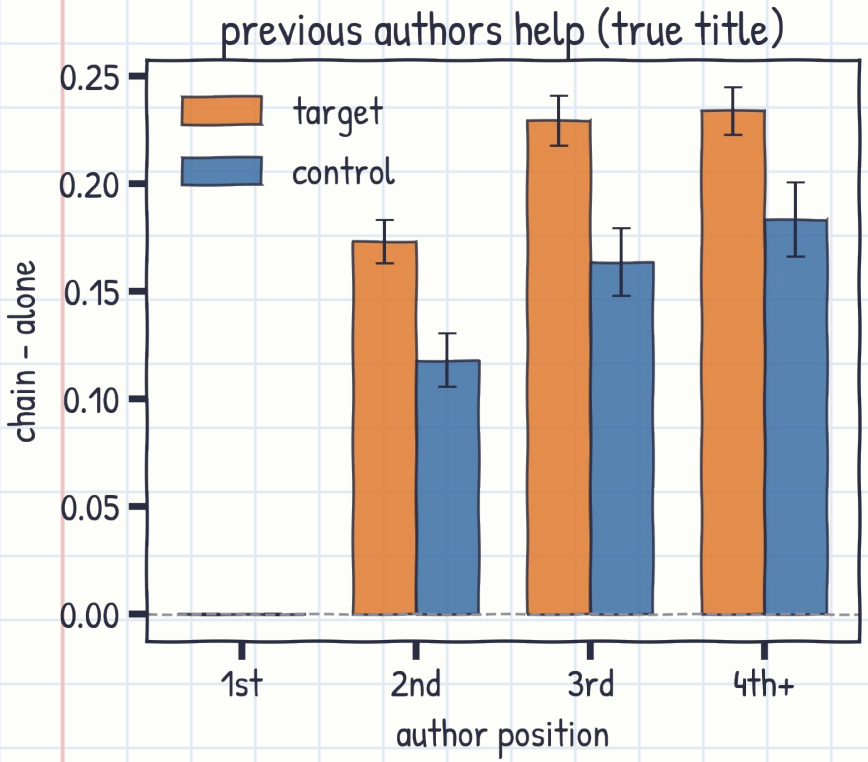

Logit lens: almost nothing until **L48**

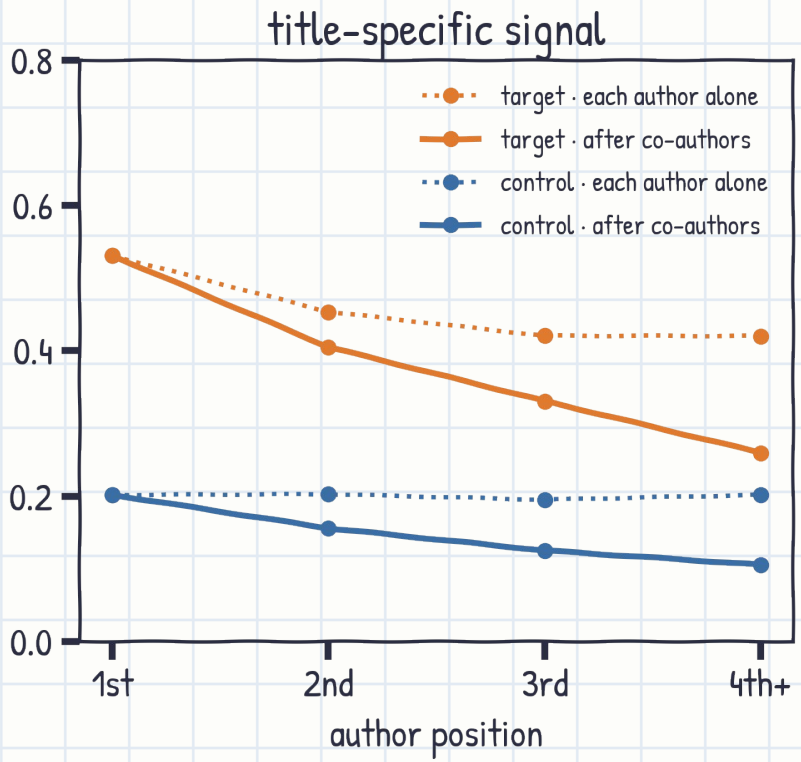
Jacobian lenses read more from **L40**

Everything jumps at **L62** (rank ~10)

# Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI



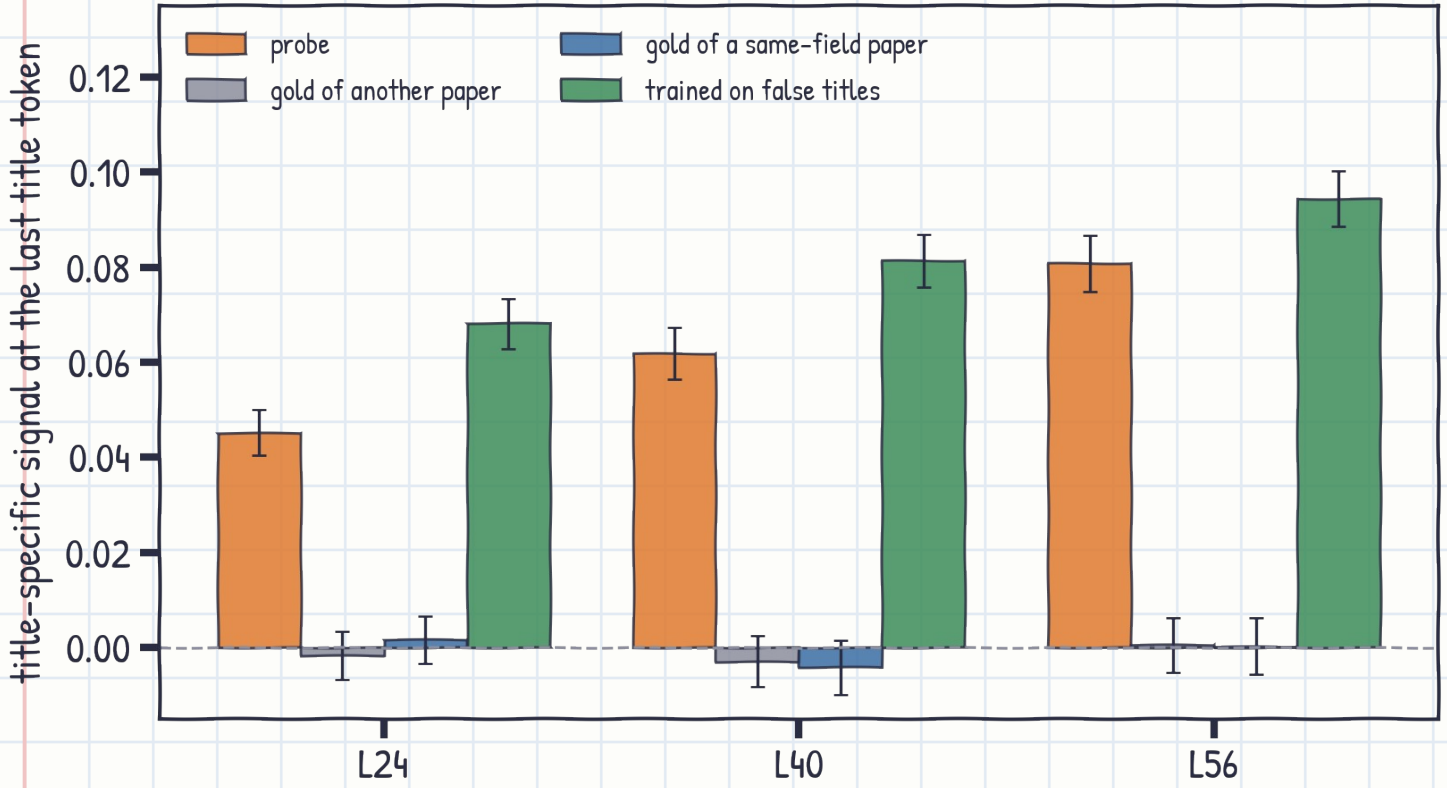

later authors +0.17 to +0.23

title-specific part **shrinks**

target > control still

# The probe signal is paper-specific

last title token · 5-fold ridge probe · controls · all 9,108 papers




another paper's authors: 0

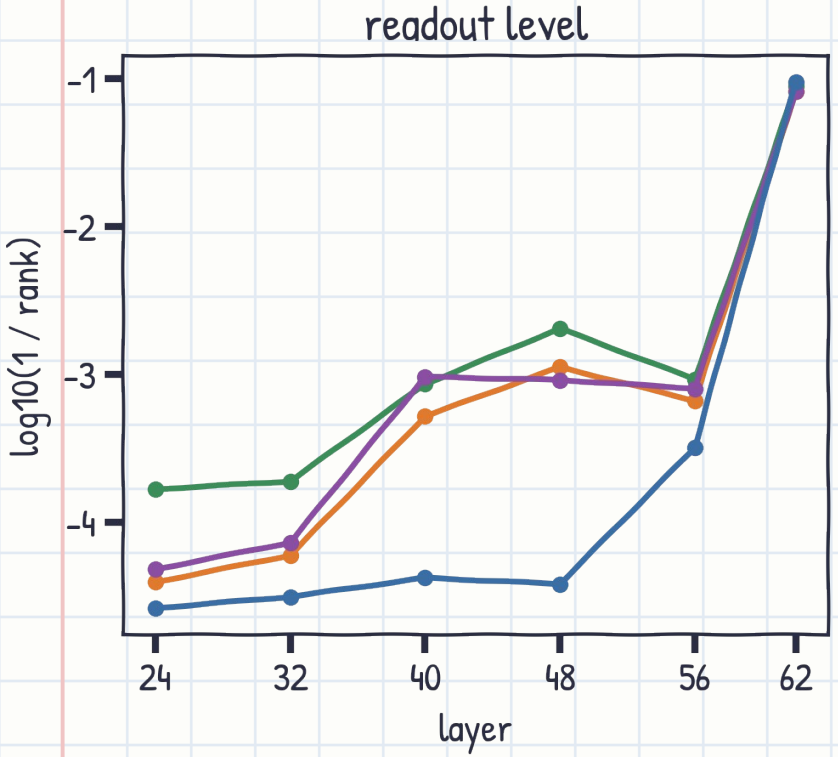
same-field paper: 0

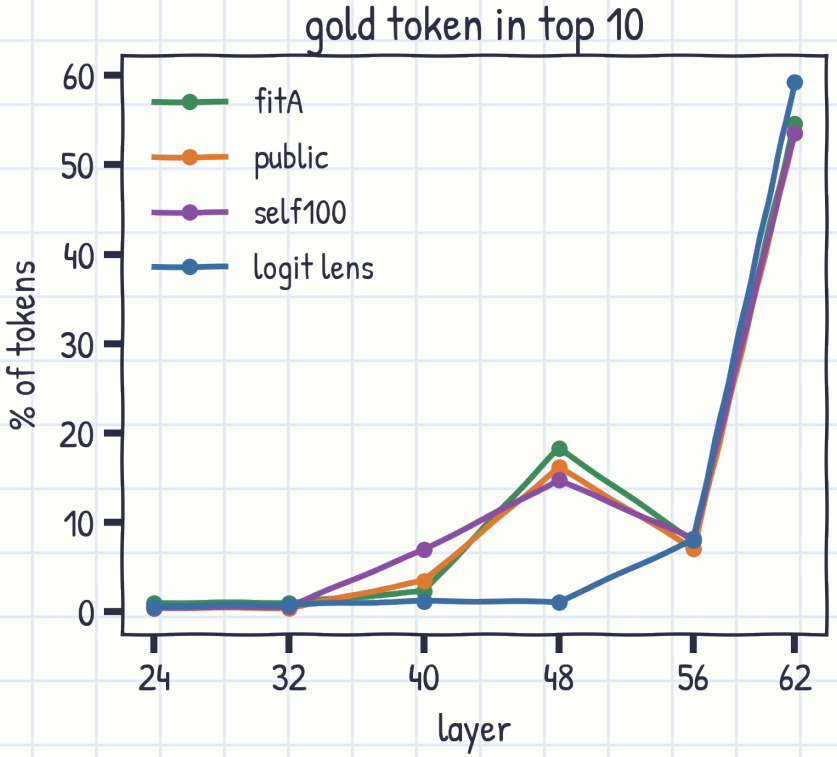
so it is this paper, already at L24

lenses cannot read it

# Which lens reads best?

all papers · true title



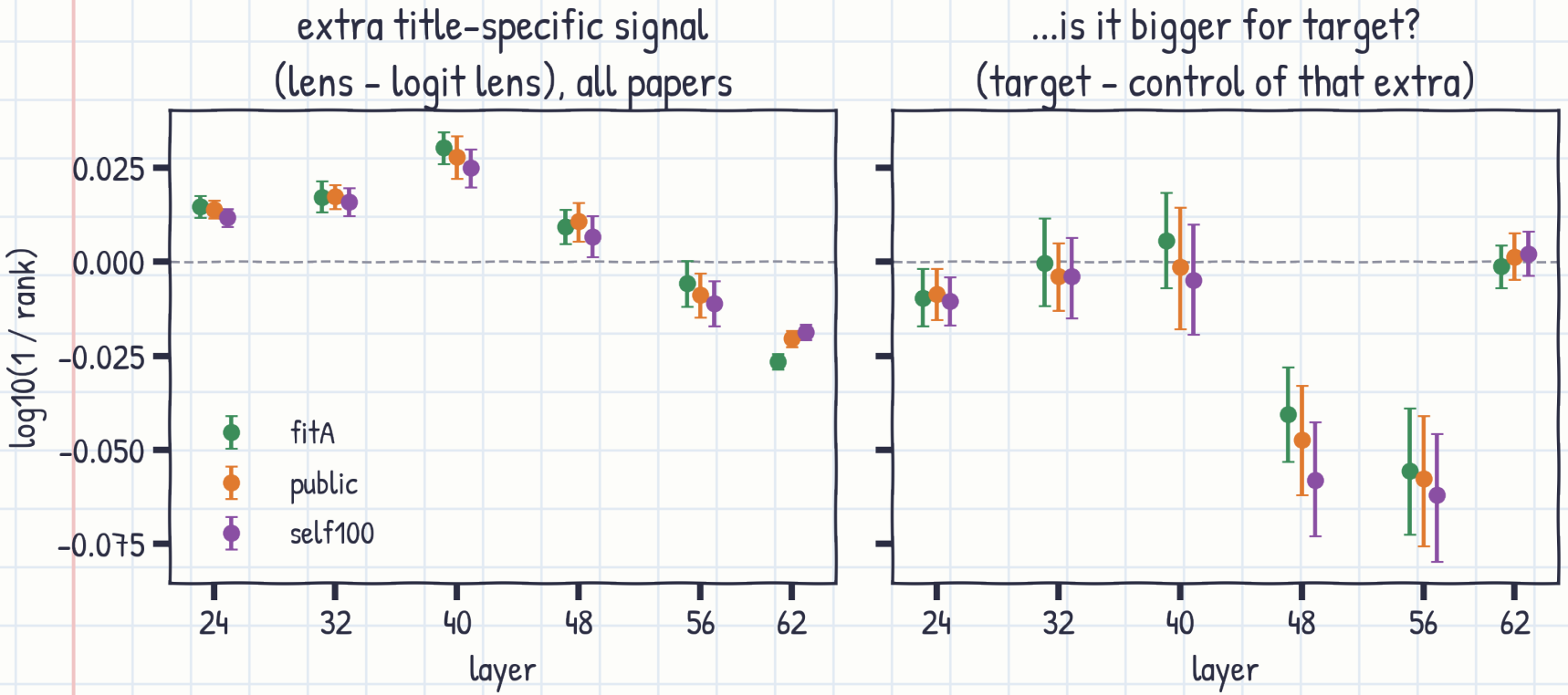

**Middle layers:**

fitA best; at L48 15–18% top-10 vs 1% for logit

**L62:** logit lens best (59%)

# Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



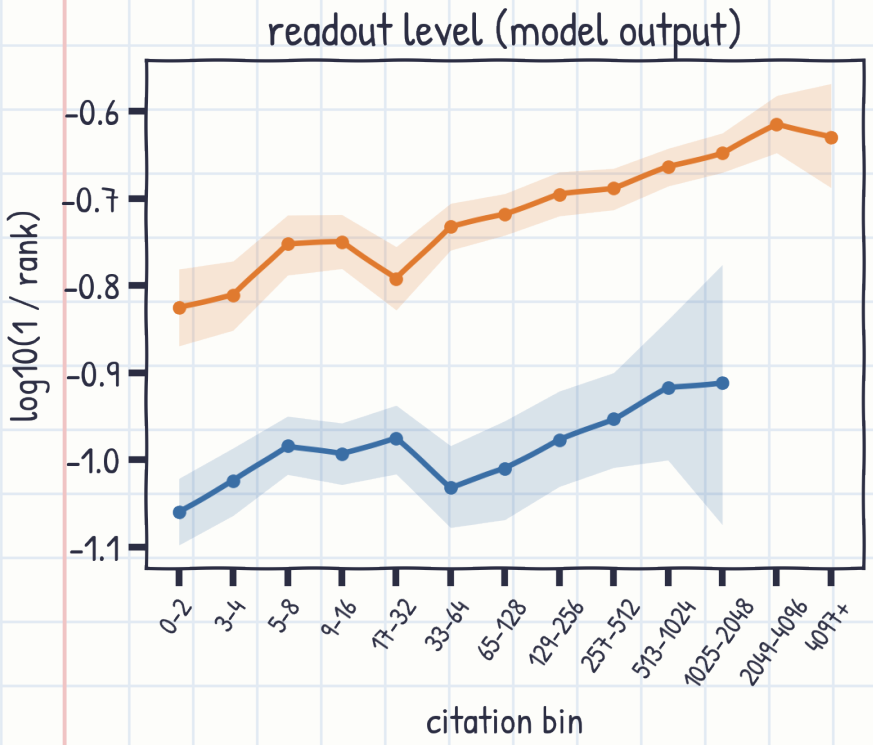

Extra signal: **+0.01 to +0.03**

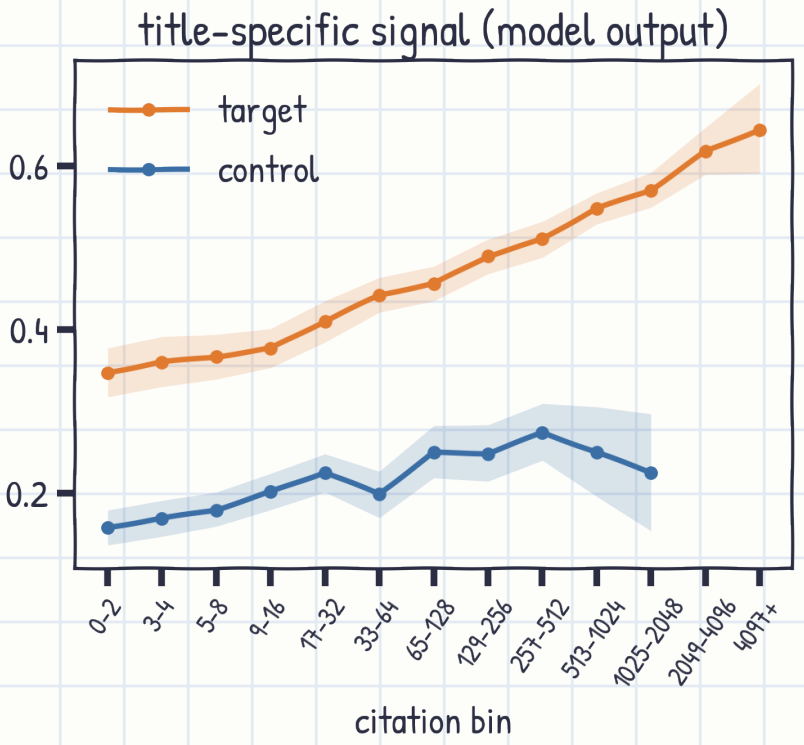
vs ~0.44 at L62

**Not bigger for target**
(even negative at L48–56)

# More citations, more paper knowledge

model output · bins with < 10 papers omitted




Target: +0.35 -> +0.64

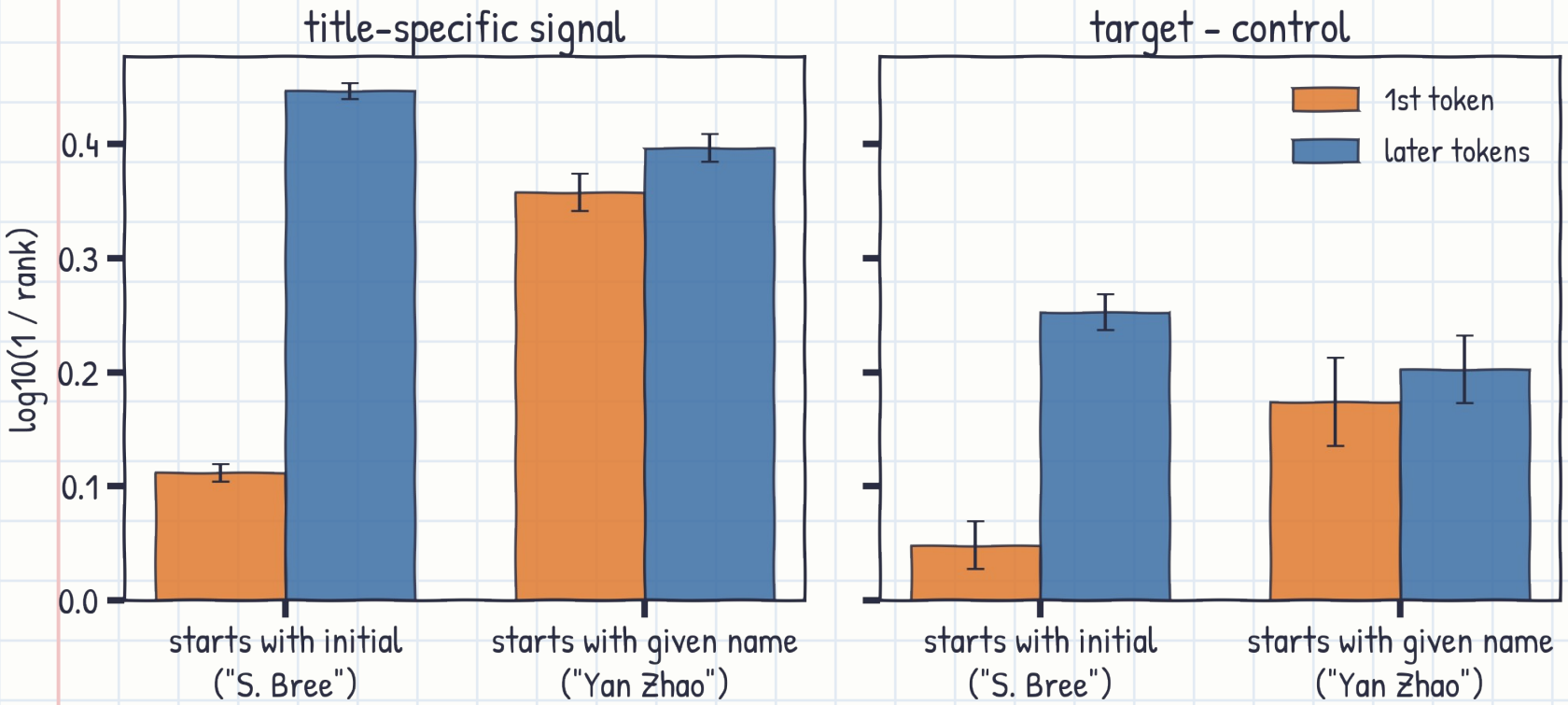
Spearman (title-specific)

**0.45** all · 0.33 target · 0.19 control

Target > control in every bin

# First token or the rest of the name?

model output · split by name format

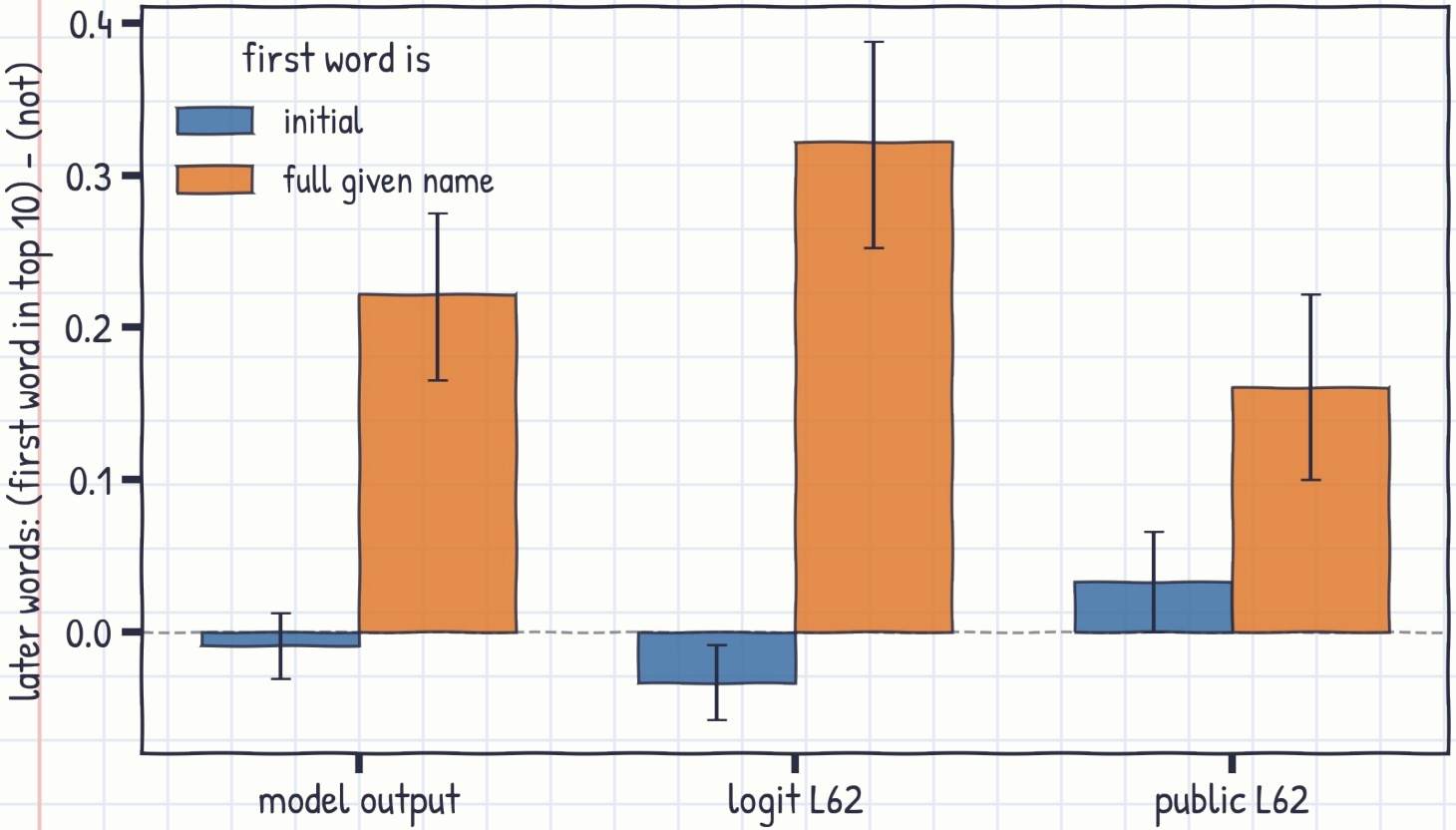



**Initial:** carries little; signal is on the surname

**Full given name:** carries about as much as the surname

# Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10




**Given name in top 10**

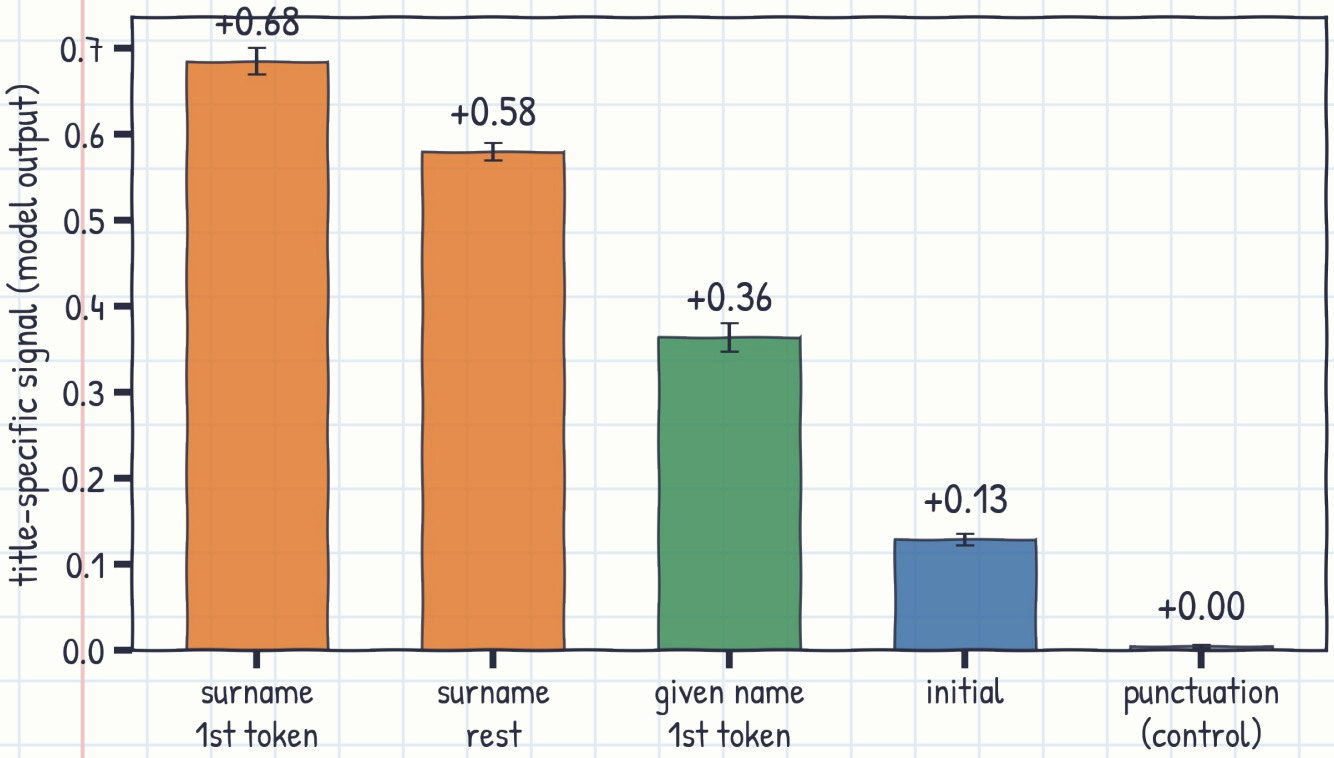
surname +0.22 (output)

+0.32 (logit L62)

**Initial in top 10**: no help

# Where the signal sits, and a dose-response

model output







Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, p < 0.001.